

Multimodal Large Language Models (MLLMs)

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Connecting Vision, Language, and Beyond



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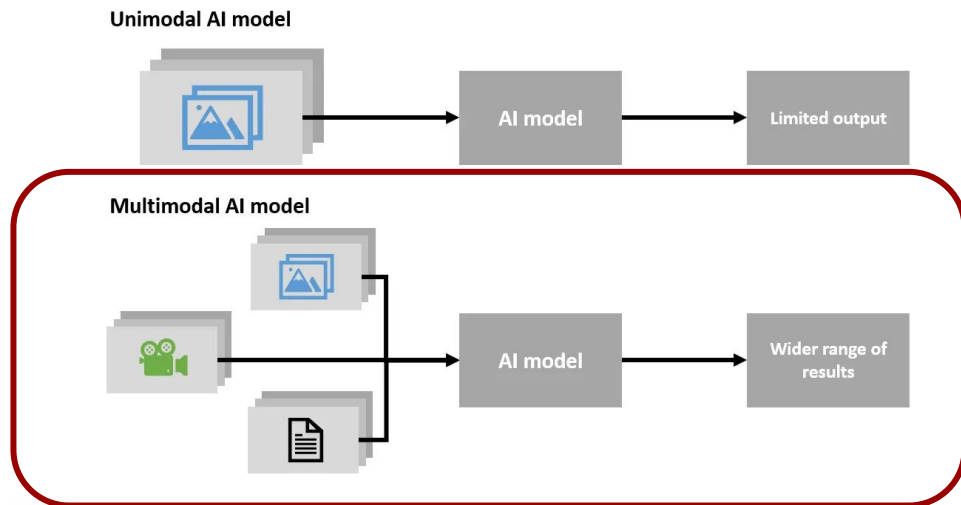
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Definition

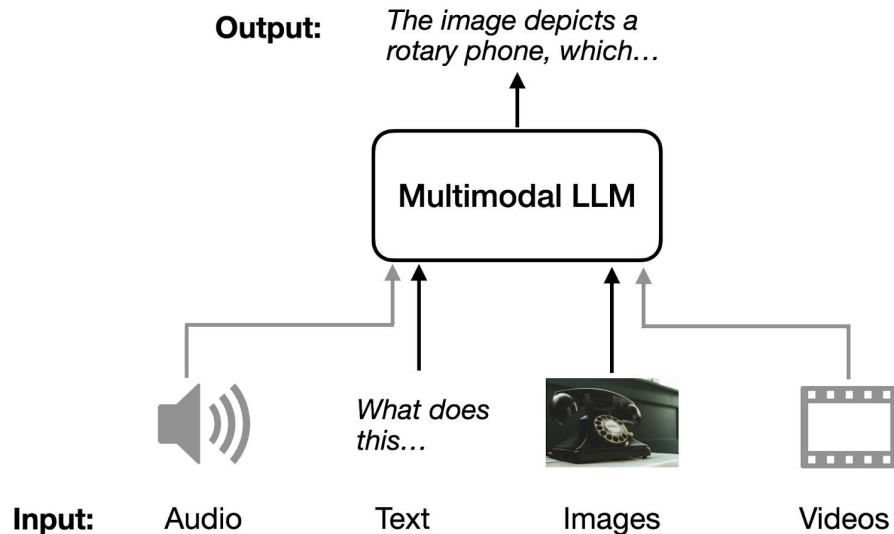
What is a Multimodal Model?

A model capable of processing and integrating information from two or more modalities (e.g., text + image).



Introduction

- An MLLM is an LLM-based model that possesses the ability to receive, reason, and output multimodal information



Motivation

- **Complexity Handling:** Real-world perception is intrinsically multimodal
 - Humans seamlessly integrate vision, language, sound, and temporal cues to understand and act in complex environments.
- **Complementarity:** MLLMs bridge the gap between Large Language Models (LLMs), which show strong zero/few-shot reasoning but are inherently "blind" to non-text inputs, and Large Vision Models (LVMs), which can "see clearly" but often lag in complex reasoning.
- **Efficiency:** MLLMs mitigate the substantial computational costs associated with training traditional multimodal models from scratch by leveraging and integrating pre-trained unimodal foundation models (frozen components).
- **Emergent Capabilities:** MLLMs inherit cognitive properties from LLMs (e.g., robust language generation, Chain of thoughts, In Context Learning) and exhibit surprising emergent capabilities, such as writing website code based on images or performing OCR-free math reasoning.



Main Challenges

- **Multimodal Hallucination:** MLLMs often generate content misaligned with the visual input
 - Existence hallucinations (describing non-existent objects).
 - Attribute hallucinations (assigning false properties).
- **Data Scarcity & Quality:** High-quality, fine-grained multimodal instruction data is expensive and limited while large-scale web data remains noisy and weakly paired.
- **Context Length & Reasoning Limits:** Current MLLMs have difficulty processing long or interleaved multimodal sequences, limiting applications such as long-video reasoning, multi-page document analysis, or temporal scene understanding.
- **Linguistic Bias & Over-Reliance:** Models tend to default to text-based priors, producing plausible and sounding, but visually incorrect answers when the image is ambiguous, revealing pattern-matching rather than genuine cross-modal understanding.

Classical Problems

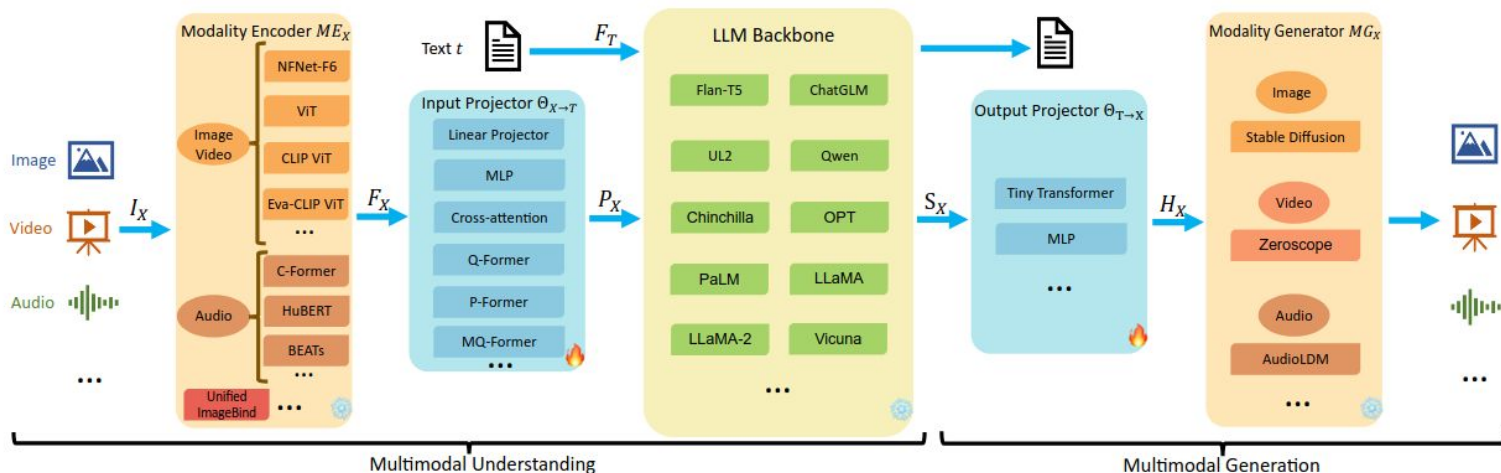
- **Visual Question Answering (VQA):** Answering natural language questions based on image or video input.
- **Image Captioning and Description:** Generating coherent textual summaries for images.
- **Document Understanding:** Interpreting charts, diagrams, and scanned documents with both text and layout reasoning.
- **Multimodal Retrieval and Search:** Matching text queries to relevant images or videos (and vice versa).
- **Video Understanding:** Summarizing or reasoning over events in video sequences.
- **Robotics and Embodied AI:** Enabling robots to understand and act on visual and textual instructions.
- **Medical Imaging Analysis:** Interpreting radiology images in combination with clinical notes for diagnosis assistance.
- **Education and Accessibility:** Explaining visual content for visually impaired users or educational feedback systems.



MLLM Architecture: Overview

Multimodal architectures usually follow a common structure that consists of:

- **Modality Encoders:** for each type of data (visual, textual, audio, etc), uses a pretrained model to extract a features.
- **Modality interface:** these are responsible for combining all the different input features into a shared one.
- **Modality Generator:** generates a desired output depending of the model task (e.g. a new image, video, text, etc).

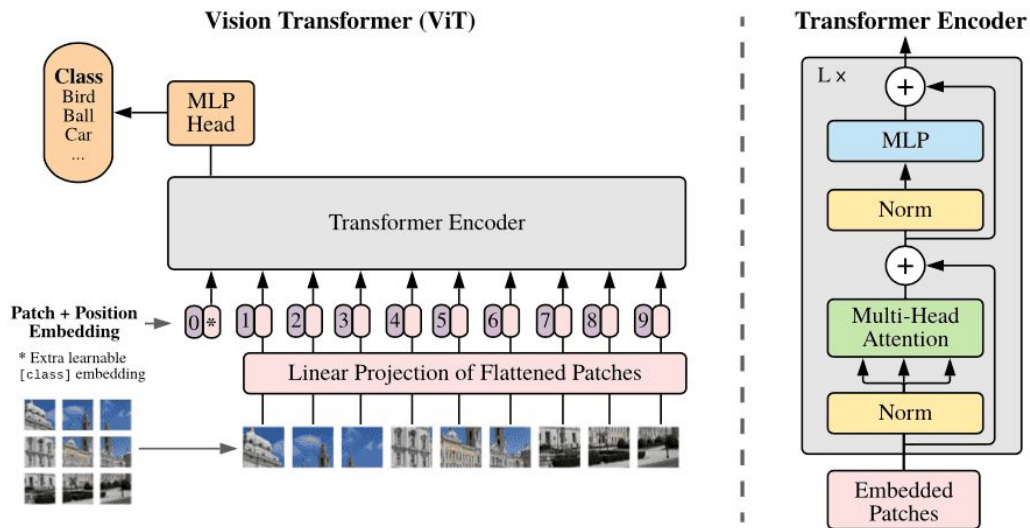


[MM-LLMs: Recent Advances in MultiModal Large Language Models](#)



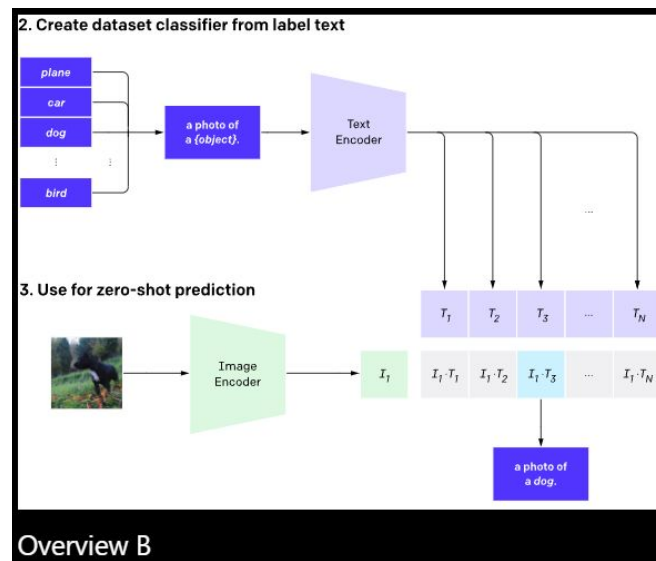
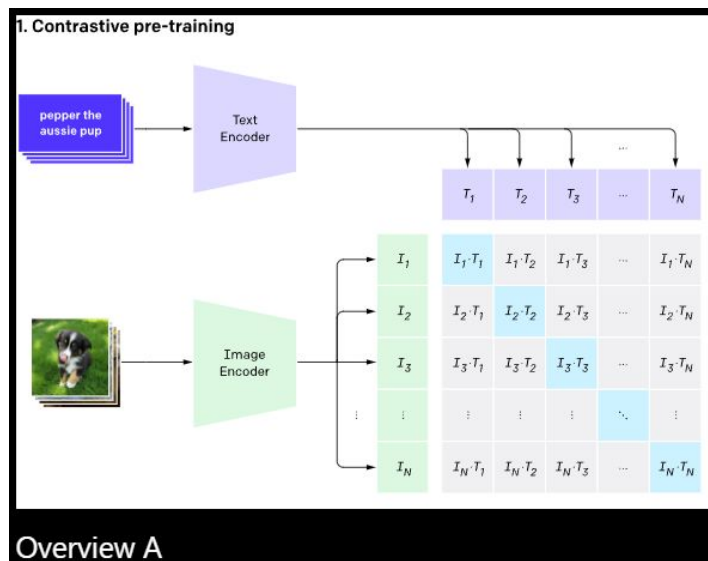
MLLM Architecture: Modality Encoder

- **Modality Encoder (ME):** Compresses raw inputs (images, audio, 3D data, etc) into compact features (F_x). Examples include specialized encoders (CLIP ViT, HuBERT) or unified encoders (ImageBind).



[An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale](#)

MLLM Architecture: Modality Encoder



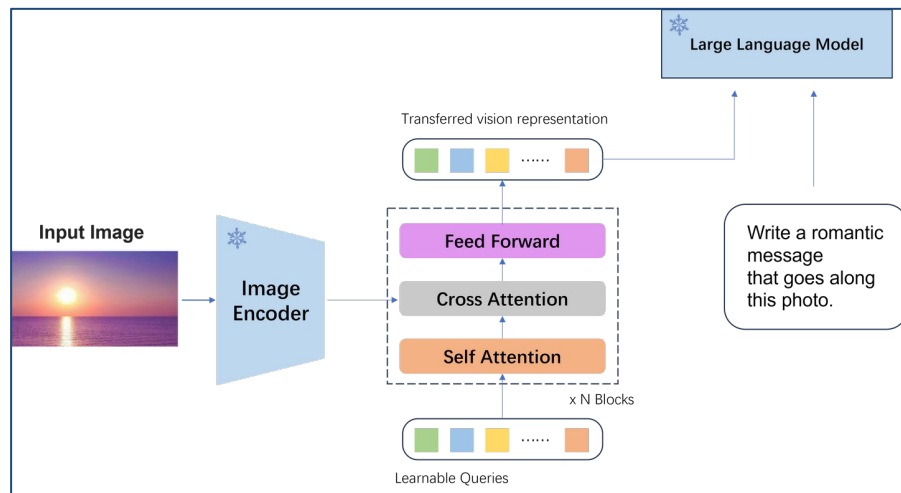
"We show that scaling a simple pre-training task is sufficient to achieve competitive zero-shot performance on a great variety of image classification datasets. Our method uses an abundantly available source of supervision: the text paired with images found across the internet. This data is used to create the following proxy training task for CLIP: given an image, predict which out of a set of 32,768 randomly sampled text snippets, was actually paired with it in our dataset. In order to solve this task, our intuition is that CLIP models will need to learn to recognize a wide variety of visual concepts in images and associate them with their names. As a result, CLIP models can then be applied to nearly arbitrary visual classification tasks. For instance, if the task of a dataset is classifying photos of dogs vs cats we check for each image whether a CLIP model predicts the text description "a photo of a dog" or "a photo of a cat" is more likely to be paired with it."

[CLIP: Connecting text and images](#)

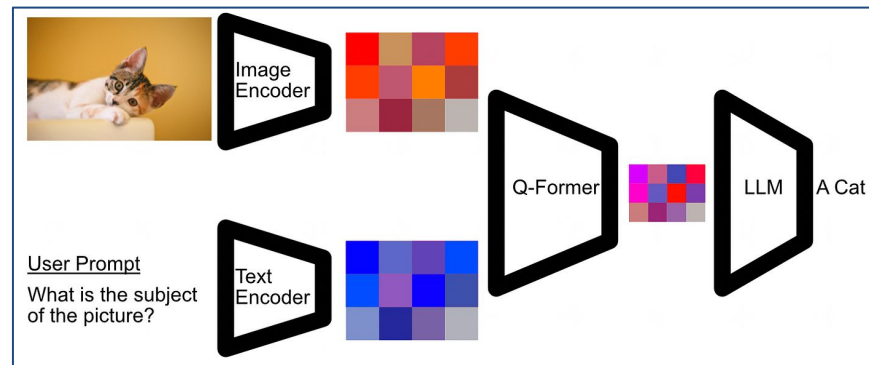


MLLM Architecture: Modality Interface

- **Modality Interface / Input Projector:** This learnable connector aligns the encoded features (F_x) with the LLM's text feature space (T) to bridge the modality gap.
 - Implementation: Typically simple structures like Linear Projectors or MLPs, or more complex query-based methods like Q-Former.



[BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models](#)



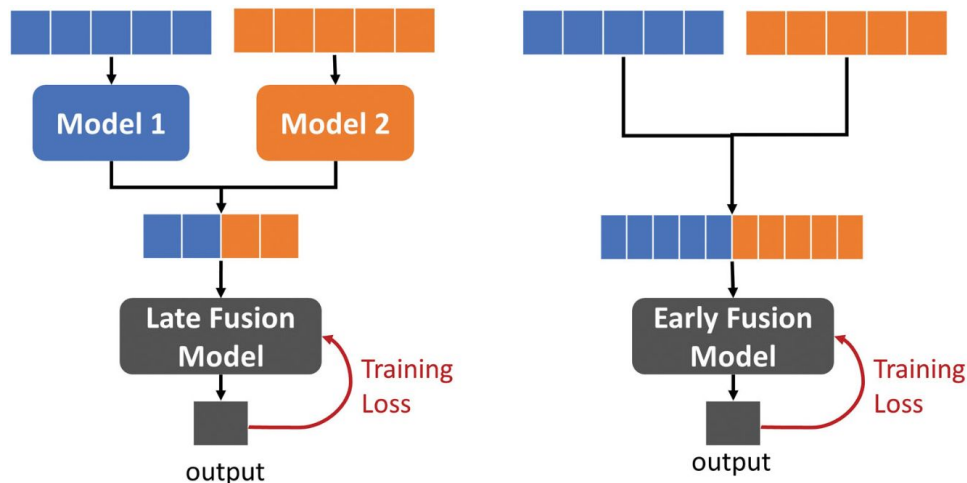
[Visual Question Answering with Frozen Large Language Models](#)

MLLM Architecture: Alignment/Fusion

Fusion: combining multimodal features into a shared representation for the LLM.

Late fusion: combine LLM outputs or logits after separate processing; simpler but may lose cross-modal interactions.

Early fusion: merge features before or inside the LLM; allows joint reasoning but can be costly.

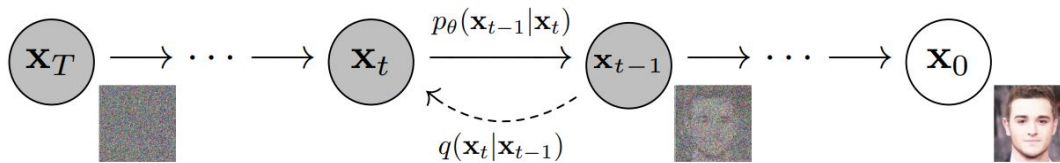


[Fusion Modeling: Combining Clinical and Imaging Data to Advance Cardiac Care](#)

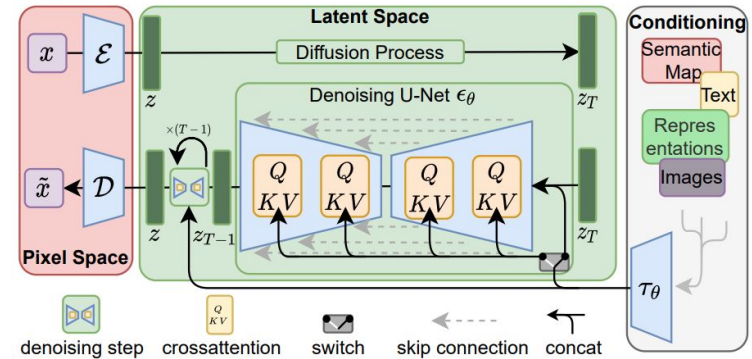


MLLM Architecture: Modality Generator

- **Stable Diffusion** is one of the modality generators commonly used by the community.
- It is an evolution from the diffusion models.
- It uses some of the principles of the Brownian motion.
 - From random small steps it converges to a coherent image.



[Denoising Diffusion Probabilistic Models](#)

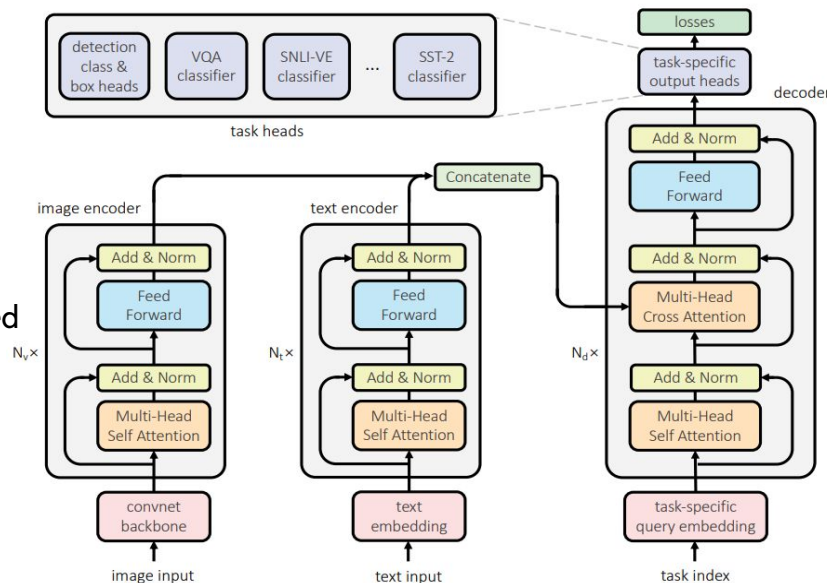


[High-Resolution Image Synthesis with Latent Diffusion Models](#)

Conclusion

- CLIP: aligned embeddings
- BLIP-2: bridge via Q-former
- Flamingo: multimodal prompting
 - The LLM itself is largely modality-agnostic; the encoders and projectors are bridging mechanisms.
- GPT-4V / Gemini: unified transformer backbone → Same model interprets text, images, diagrams, and charts
 - A single model backbone directly accepts multiple modalities as inputs. No separate encoders/projectors are needed for each modality; the model learns a shared representation space internally.

Trend: Moving from “fusion” to integration



Future Directions

- **Deeper Integration over Simple Fusion:** The field is moving away from shallow feature fusion toward architectures that learn unified, joint representations across modalities. This enables more flexible, context-aware models that generalize better.
- **Physical AI and Embodied Intelligence:** Multimodal models are increasingly applied to real-world systems, e.g., robots, AR/VR, where inputs like IMU, depth, tactile, and audio are used to perceive and act in the physical environment.
- **Any-to-Any Modality Generalization:** Emerging models are designed to map flexibly across input-output pairs, e.g., text-to-audio, image-to-action, rather than being locked into fixed modality pairs. This supports open-ended, zero-shot reasoning.
- **Unified Foundation Models:** Building on CLIP, ImageBind, and beyond, we're seeing general-purpose models that can handle many modalities within a single backbone.



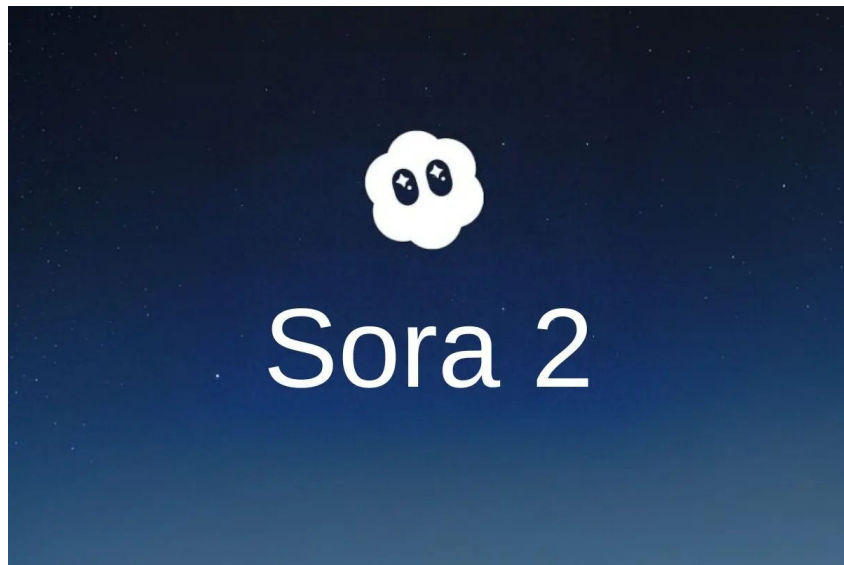
Applications



[Photoshop Generative Fill - from photoshop training channel](#)



Applications

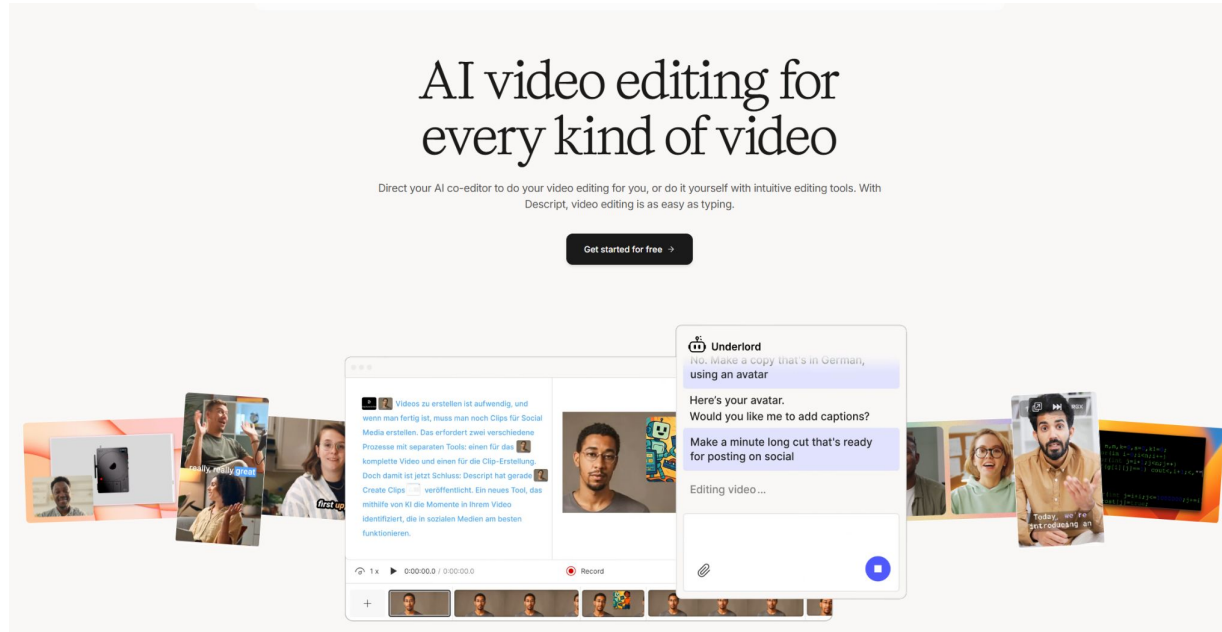


[OpenAI - Sora 2](#)



[Google - VEO 3](#)

Applications



[Descript](https://www.descript.com/)



Applications



[Med-PaLM](#)



[Project Astra](#)

Thank you!

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